

Population turnover and churn: enhancing understanding of internal migration in Britain through measures of stability

Adam Dennett and John Stillwell
University of Leeds

Net migration measures take account of the direction of migration flows, but our understanding of migration can be extended using population turnover and churn as measures of population stability. Turnover is a measure of the intensity of migration into and out of a district, whereas churn incorporates these flows and also includes the flows that take place within each district. Using districts of Britain and their type-based groupings, the highest levels of turnover and churn are found in London and some of the more dynamic urban areas, whereas the lowest levels are found in rural and previously industrial areas. Age has a significant effect on these measures with the population in their late teens and early twenties being the least stable and older populations being more stable.

Introduction

In demography, population stability is usually associated with constant fertility and mortality, such that there are neither increases in size nor changes in age structure. There is a body of stable population theory that provides a mathematical framework for connecting a fixed set of rates to the population dynamics that they generate. However, the authors must acknowledge that population size and structure are significantly influenced by the flows of international migration at the national level and also by flows of internal migrants at the subnational scale. In this paper, our concern is with the latter component – internal migration – and the use of measures of ‘turnover’ and ‘churn’ as indicators of population stability at the district scale in Great Britain. Turnover is a measure of the intensity of migration into and out of a district, whereas churn incorporates these flows but also includes the flows that take place within each district. While net migration balances and rates summarise the relationship between inflows and outflows and provide an indication of population dynamics, measures of population turnover and churn can also be used in conjunction with net migration to extend our understanding of migration by providing insights into the stability of the population at the district level. While population turnover and churn may be indicative of growth and buoyancy, the work by Rotolo and Tittle (2006) has indicated that the stability of an area’s population may be an important influencing factor for social maladies such as crime or deviant behaviour – or indeed other more positive socially-related phenomena such as general happiness. Recent work by Bailey and Livingston (2007) has provided the first systematic analysis of migration turnover and deprivation at the neighbourhood scale across Britain.

Traditionally, spatial analyses of internal migration have concentrated on the volume and direction of movement between areas at different geographical scales. Recent studies, such as those by Champion

(2005; 2007) and Dennett and Stillwell (2008), have tended to use net migration as the variable of interest and have identified significant flows of people between areas defined as ‘urban’ and ‘rural’ (counterurbanisation) or ‘London’ and the ‘rest of the UK’. In this paper, inter-district net migration is used as the context for investigating how turnover and churn vary spatially across the country and the relationship between these measures and net migration is examined. In order to facilitate the interpretation of large quantities of data, a district classification system is used to identify features and patterns. The purpose of the paper is therefore to show how these measures of population stability enhance our understanding of internal migration in Britain. However, before reporting on the analysis, we begin with a short introduction to the data used in the paper and the classification system that has been adopted.

Data sources

The data used in this analysis are from the 2001 Special Migration Statistics (SMS), a source of counts of migrant flow between origins and destinations during the 12-month period before the 2001 Census. SMS are available at three spatial levels: districts, wards and output areas, and the flows used in this paper were obtained from SMS tables for UK districts (London boroughs, unitary authorities, metropolitan districts and other local authority districts in England; unitary authorities in Wales; council areas in Scotland; and parliamentary constituencies in Northern Ireland) (Stillwell and Duke-Williams, 2007). The data were extracted using the Web-based Interface to Census Interaction Data (WICID), the query-based software system that allows researchers in the UK to access the census interaction data online (Stillwell and Duke-Williams, 2003). The analysis which follows uses data for Great Britain and not for the whole of the UK. Northern Ireland has been omitted from the analysis for a number of practical, data-related reasons documented in Dennett and Stillwell (2008). The framework for the analysis, is a national district level three-tier classification developed by Vickers et al. (2003) in collaboration with ONS, which distinguishes four ‘families’ of districts at the most aggregate level (**Figure 1**), 24 ‘classes’ at the most disaggregate level and an intermediate set of 12 ‘groups’. The classification, derived using Ward’s hierarchical grouping procedure based on the use of 56 variables, serves two purposes: firstly, it allows the reduction of large interaction matrices of 408×408 origins and destinations in Britain to a more manageable combination of 4×4 , 12×12 and 24×24 matrices. Secondly, the classification has been created to group and differentiate areas by certain key socio-demographic and socio-economic characteristics. Studying migration flows between areas grouped according to these underlying characteristics reveals more about the associations between particular types of area and specific migration flows.

Results

Net migration

To provide context for the measurements of population turnover and churn, it is useful to summarise the aggregate net migration balances and rates for Great Britain for each of the ‘families’, ‘groups’ and ‘classes’ in the Vickers et al. classification. **Table 1** is a summary at each of the three levels of the net migration characteristics of districts whose net migration rates are mapped in **Figure 2**. The authors have discussed these patterns in detail elsewhere (Dennett and Stillwell, 2008), so here we give a short overview of the key features.

Figure 2 shows that patterns of net gain and loss for migrants of all ages tend to be associated with areas generally recognisable as rural and urban respectively, as discussed by Champion (2005). Most districts of Greater London and those bordering (including those stretching out along the M4) are experiencing net out-migration. Other metropolitan areas, including Birmingham, Liverpool, Manchester and their surrounding

Table 1

Net migrants and net migration rates by district classification – all people, Great Britain, 2000–01

District classification (Family, Group, Class)	Net migrants	Net migration rate (per 1,000 population)
A: Urban UK	-4,842	-0.23
A1/A1a: Industrial Legacy	-5,263	-0.95
A2: Established Urban Centres	-16,829	-1.69
A2a: Struggling Urban Manufacturing	-8,292	-2.78
A2b: Regional Centres	3,195	1.96
A2c: Multicultural England	-11,264	-3.13
A2d: M8 Corridor	-468	-0.26
A3: Young and Vibrant Cities	17,250	3.38
A3a: Redeveloping Urban Centres	15,395	3.89
A3b: Young Multicultural	1,855	1.61
B: Rural UK	57,947	2.72
B1: Rural Britain	30,653	3.55
B1a: Rural Extremes	1,067	0.67
B1b: Agricultural Fringe	14,762	4.32
B1c: Rural Fringe	14,824	4.09
B2: Coastal Britain	29,296	6.59
B2a: Coastal Resorts	7,231	7.27
B2b: Aged Coastal Extremities	13,987	5.20
B2c: Aged Coastal Resorts	8,078	10.62
B3: Averageville	-2,040	-0.25
B3a: Mixed Urban	-2,311	-0.45
B3b: Typical Towns	271	0.09
B4/B4a: Isles of Scilly	38	17.79
C: Prosperous Britain	-4,983	-0.52
C1: Prosperous Urbanities	1,844	0.58
C1a: Historic Cities	5,606	3.48
C1b: Thriving Outer London	-3,762	-2.41
C2/C2a: Commuter Belt	-6,827	-1.07
D: Urban London	-48,122	-8.53
D1/D1a: Multicultural Outer London	-20,947	-8.01
D2: Mercantile Inner London	-11,705	-10.21
D2a: Central London	-11,725	-10.29
D2b: City of London	20	2.79
D3: Cosmopolitan Inner London	-15,470	-8.22
D3a: Afro-Caribbean Ethnic Borough	-7,164	-6.07
D3b: Multicultural Inner London	-8,306	-11.81

districts, the North East and Glasgow, have negative net migration rates. In contrast, more rural areas covering large parts of East Anglia, the South West, Wales, the Midlands, the North and Scotland are all experiencing rates of net in-migration.

The patterns revealed on the map (Figure 2) are summarised at the most aggregate family level in the classification (Table 1), with Urban UK, Urban London and Prosperous Britain exhibiting net out-migration, and Rural UK exhibiting net in-migration, the highest rates of movement being for Urban London and Rural UK. Closer inspection of the groups and classes within each family reveals that while Urban UK as a whole is losing migrants, some areas within Urban UK are gaining significant numbers of migrants and exhibiting positive net migration rates. Drilling down through the classification hierarchy reveals other examples of net inflows or outflows masked at the broad family level: Averageville within Rural UK, Historic Cities within Prosperous Britain and City of London within Urban London all exhibit the opposite net migration balance to their parent family.

Migration by age and district type

But the study of net migration balances and rates is not the complete story. Net migration is useful for understanding the actual and relative

Figure 1

Districts in Great Britain as classified by Vickers *et al.* (2003)

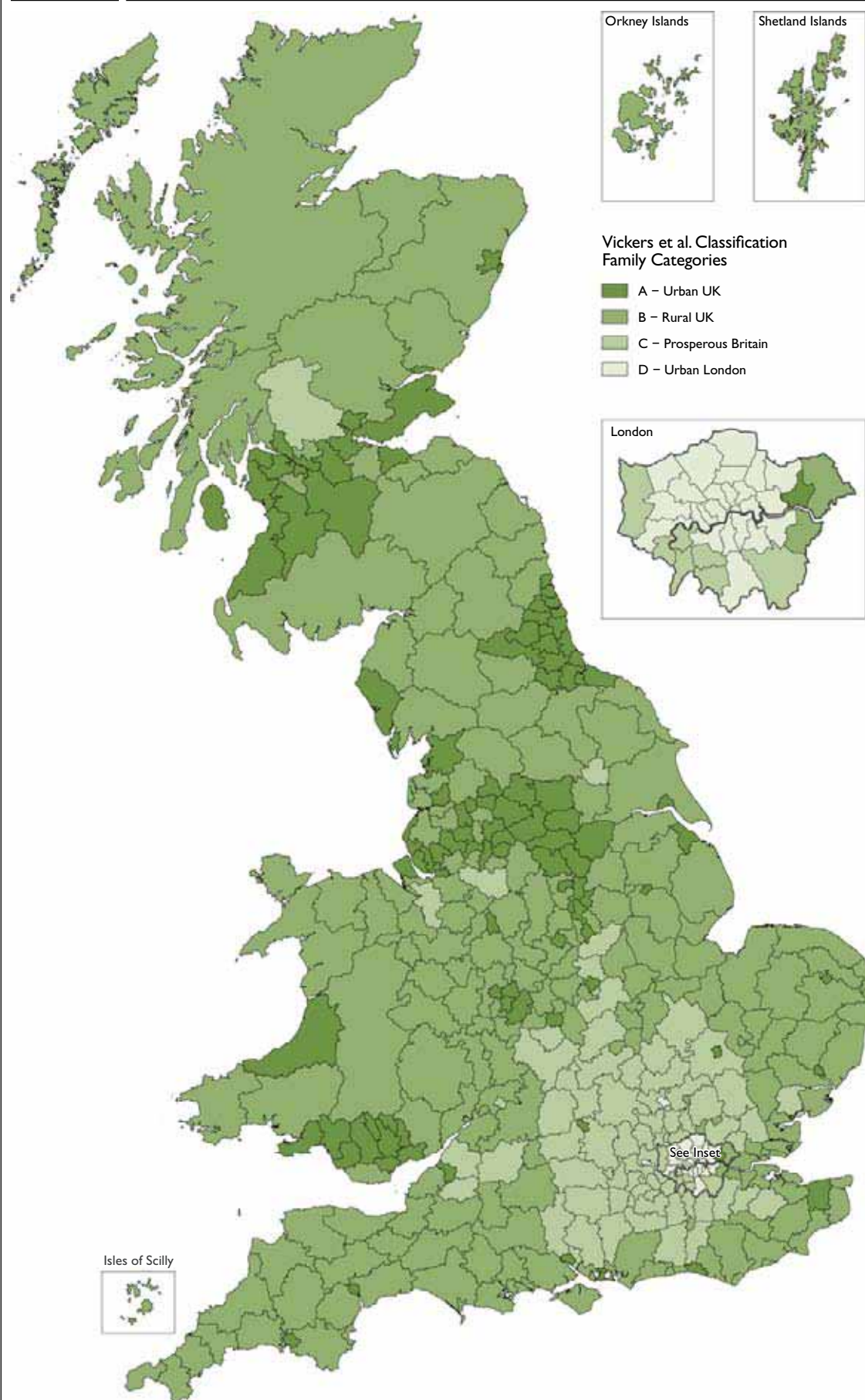


Figure 2

Great Britain: District net migration rates for all ages, 2000–01

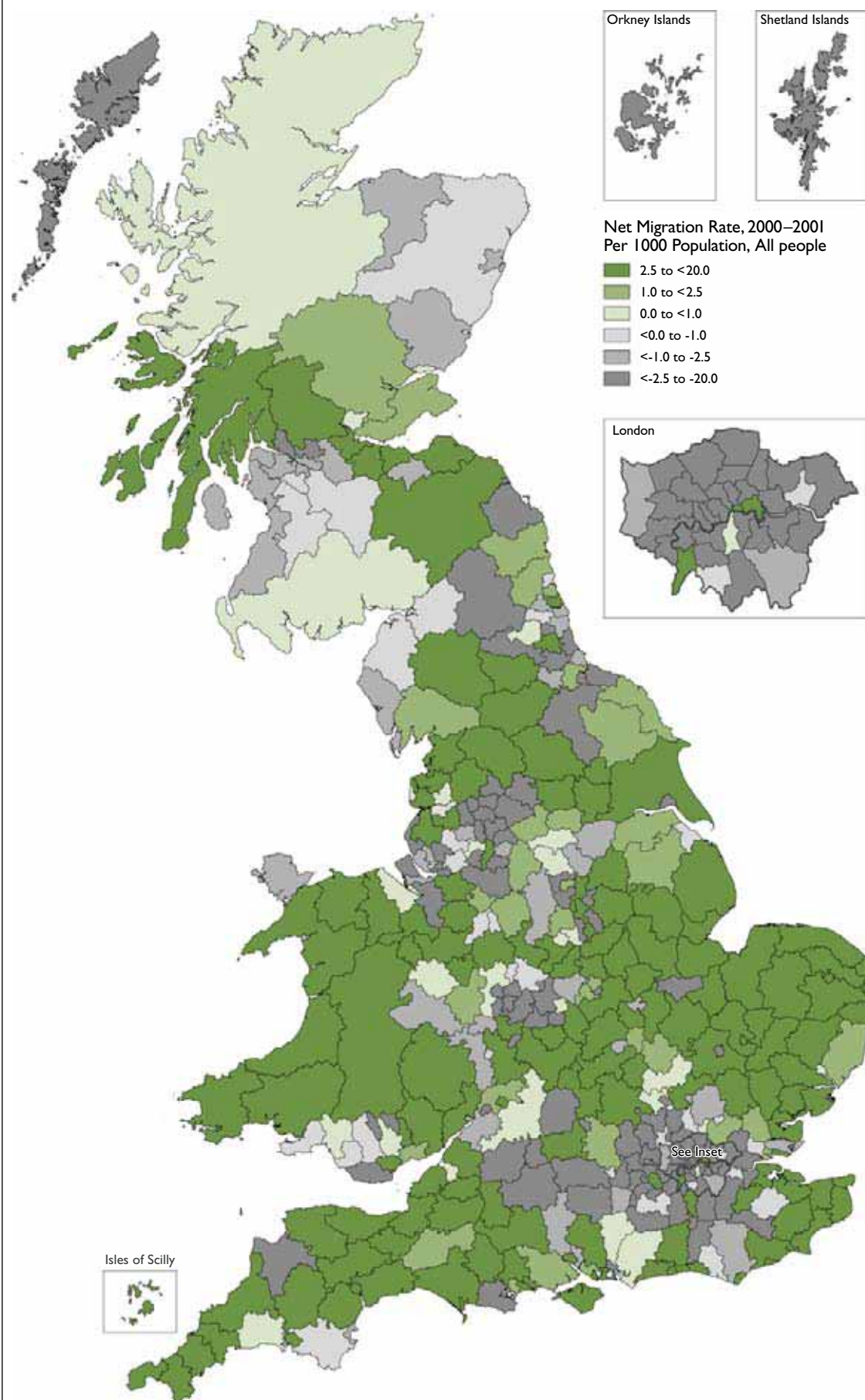
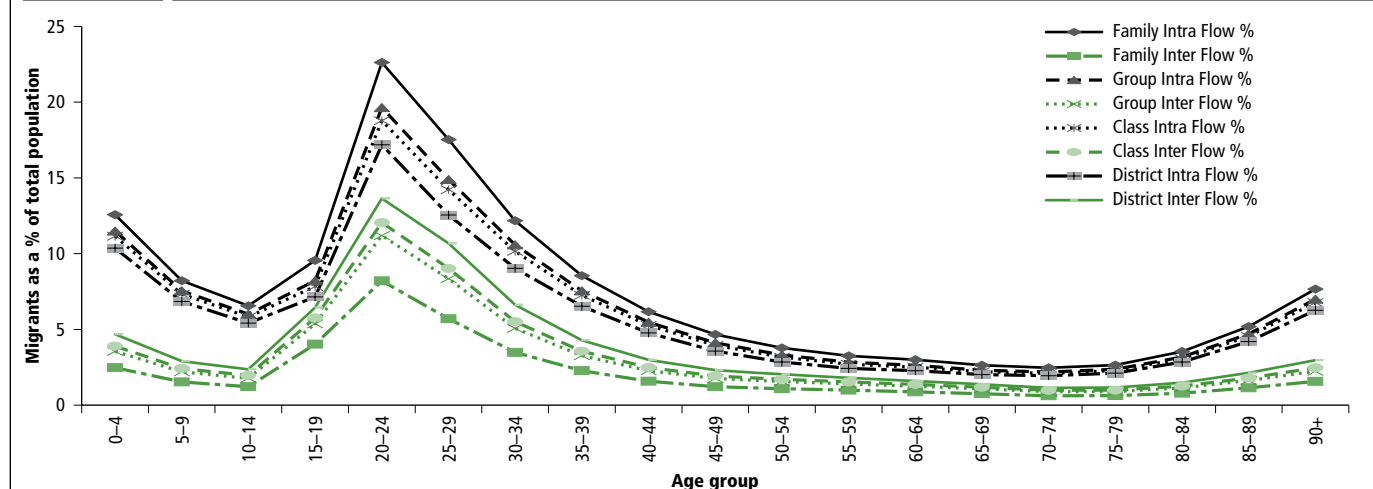


Figure 3

Internal migration age schedules by district type and type of flow, Great Britain, 2000–01



movements of groups between areas but – as was outlined by Ravenstein (1885; 1889) in his seminal work on migration, and as has also been noted frequently by subsequent works on migration over the years including Westefeld (1940), Zipf (1946) and Lawton (1968) – distance plays a very important role in migration. Shorter-distance moves are far more common and frequent than longer-distance moves, particularly in densely populated urban areas. In the context of this study the spatial scale is that of the local authority district. The average area of the districts in Great Britain is 567 sq km. Since districts cover relatively large units of space, moves between districts are likely to be of a longer distance, and thus in all probability less frequent than moves within districts. These intra-zone moves are of even more significance when the district classification hierarchy used in this analysis is taken into consideration. Classes, groups and families of district create successive geographical areas of increasing size, such that intra-zonal moves also increase in importance at each level. This relationship is confirmed by examining the intra/inter zonal migration schedules shown in **Figure 3**. Throughout the life course, at each age group, intra-zonal flows are of more importance than inter-zonal flows. Furthermore, the gap in importance increases when moving up through the classification hierarchy from the basic district building block. At any age, the gap is smallest between the intra-district and inter-district flows, whereas the gap is largest between intra-family and inter-family. By only studying net migration flows and rates between districts and classifications of district, and not including the flows within areas, we can be certain that much of the story of migration within Britain is being overlooked.

Measuring turnover and churn

The net migration variables (balances and rates) used so far provide some measure of the stability of the population. Of course, natural change components such as births and deaths will have an impact as well as international migration, and certainly there will be some areas that feature above or below average birth, death or external migration rates. However, internal migration has been the most important component driving population change across Britain in many areas in recent decades (Rees et al., 1996; Christophersen, 1997), although the impact of international migration in the South (Champion and Congdon, 1992) has been of considerable importance since the 1980s, with London being the most common destination in Great Britain for immigrants in the 12-month period before the 2001 Census (Horsfield, 2005). Since 2001, while there have been significant inflows from overseas of asylum seekers to London and new labour migrants to other parts of the country, as well as steady flows of emigrants, the average international turnover of population in England and Wales has only been around 800,000

people per year since 2001. On the other hand the volume of inter-district migration in England and Wales, as indicated by patient registration data, has remained at over 2.4 million per year (ONS, 2008).

It might be assumed that a low level of net in-migration or out-migration means a relatively stable population, but this is not necessarily the case. For example, a hypothetical area with 1 million residents at the end of a year that had seen 100 residents move into the area and 101 residents depart over that period, would have a net migration rate per 1,000 people of -0.001 . This rate would be identical if, for the same area, 10,000 residents had moved in and 10,001 residents had moved out. In this example an identical rate is obscuring a hugely different turnover of population for the area and a massive change in the composition of the resident population. The limitations of net migration as a measure have been recognised (Rogers, 1989) and one alternative to measuring net migration has been to measure migration effectiveness or efficiency. This has been used in previous research both as an alternative to and in conjunction with net migration rates (Stillwell et al., 2000; 2001). Indices of migration effectiveness standardise rates of migration by using gross in-migration plus out-migration flows as the denominator rather than populations at risk (PAR). The direction of flow is standardised by the magnitude of the flows in both directions rather than the population of an area but, in doing so, the direction or symmetry of the flow is still of central importance. Consequently, the nature of the migration effectiveness measure does not make it the most suitable option for assessing the relative stability of underlying area populations.

These issues are addressed by utilising the concepts of population ‘turnover’ and ‘churn’. Population turnover (TO) by age group ‘a’ and sex ‘s’ for area ‘i’ is defined here as:

$$TO_i^{as} = \left(\frac{D_i^{as} + O_i^{as}}{P_i^{as}} \right) 1000 \quad (1)$$

where:

D_i^{as} is the in-migration of those in age group a and sex s to area i;

O_i^{as} is the out-migration of those in age group a and sex s from area i; and

P_i^{as} is the population in age group a and sex s to area i.

Turnover is useful as it takes account of both the inflow and outflow and gives a measure of how the population of an area has changed in a way that standard rate calculations (inflow minus outflow over population) do

not. It should be noted that while the term ‘turnover’ is one that can be found throughout the demographic literature, its precise meaning and the way in which it is calculated is not universal. For example, in an article published by ONS (ONS, 2007), turnover (for small areas) is calculated by averaging internal migration flows over a three-year period. This is to avoid the possible distorting effects that might be caused in smaller areas by localised phenomena, such as the building of a new housing estate or the demolition of old housing. In other work by Large and Ghosh (2006) turnover is calculated over one year but using both internal and international migration data. Bailey and Livingston (2007), on the other hand, calculate turnover just from internal migration data but, rather than using only inflows and outflows, also incorporate within-area moves. Here, we prefer to use the term population ‘churn’ (CH) for a measure that includes within-area migration as well as inflows and outflows:

$$CH_i^{as} = \left(\frac{D_i^{as} + O_i^{as} + W_i^{as}}{P_i^{as}} \right) 1000 \quad (2)$$

OR

$$CH_i^{as} = \left(\frac{D_i^{as} + O_i^{as} + W_i^{as} + W_i^{as}}{P_i^{as}} \right) 1000 \quad (3)$$

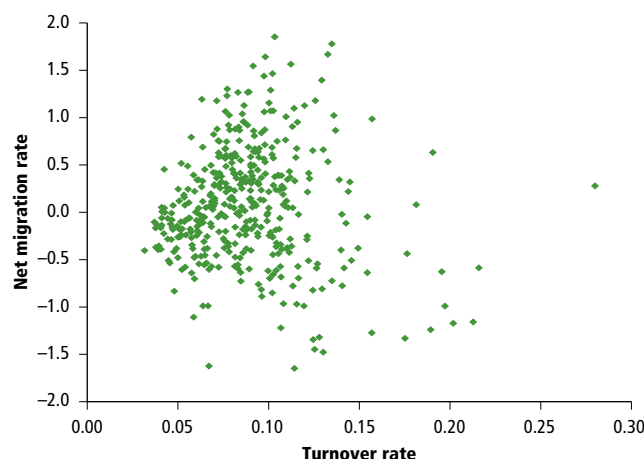
where: W_i^{as} = total migrants of age group a and sex s within area i .

Equation 2 derives a measure of churn by counting migration to an area, from an area and within an area as single events. It could be argued, however, that a within-area migration actually encompasses two migration events – leaving the origin and arriving at the destination and therefore, consistent with the inflow and outflow events being counted twice, the within-area migration event should be treated similarly, as indicated in equation 3 (the method used by Bailey and Livingston in calculating their turnover statistics). Providing two methods for measuring churn inevitably invites the question: which is the appropriate measure to use? This would depend upon the reason for measuring churn. It could be that a method for accurate measurement of the total migration **events** in an area is required; in which case, equation 3 may be more appropriate. If the reason for measuring churn, however, is to get some purchase on the stability of the population in the given area of interest (that is, how much the area is comprised of the same or different individuals between years) then counting the within-area move as two events could be more problematic, and may be misleading since it double counts a single event. If an individual leaves one residence in an area to move to another residence in the same area, then the area actually comprises the same individuals at the beginning of the period as it does at the end; the population size remains stable even though people are moving around.

The movement of an individual or individuals is likely to affect the areas they are moving from or to if the origins and destinations differ in demographic composition. If we accept Tobler’s ‘first law’ (Tobler, 1970) then closer places are more related than distant places; moves within smaller areas, therefore, are more likely to be between similar places with similar populations than moves within larger areas. Of course, as we are only using small and large areas as a proxy for distance, it is difficult to be completely accurate. While broadly accepting this, it could be said that within-area moves are less likely to be perturbing to the population of small areas than large areas, therefore, counting a within-area move as two events could be more distorting for small areas than for large. The difficulty of ascertaining the exact effect that counting within-area moves twice will have in relation to the size of the area, and in order to examine population stability defined by the extent to which populations in an area differ or remain the same over a period of time, it is more reliable to count within-area moves once. A within-area move will have some perturbing effect, and counting the migrant once acknowledges this

Figure 4

Net migration rate versus turnover rate, districts in Great Britain, 2000–01



without running into problems of possible over-emphasis. We therefore adopt equation 2 in this analysis, although it should be acknowledged that if equation 3 were adopted the interpretation of some results may differ.

As outlined by Bailey and Livingston (2007), churn is a particularly important aspect of population flow as it is associated more closely with deprivation, especially when small areas are involved. Specific local factors which may agitate local populations at the small scale cease to be important at larger scales. Despite this, measuring population churn even at coarser geographies is important for ascertaining a more accurate measurement of the relative stability of the population in different areas. It could be argued that, where two areas with the same levels of population turnover are compared, it would be the area with the higher levels of internal movement relative to the population size that would have the less stable population.

Figures 4 and 5 plot rates of net migration against turnover and churn respectively, for all districts in Britain. For both turnover and churn there is a relatively even distribution between positive and negative net migration values. However, in both cases there is a positive skew with the frequency of districts with higher rates of turnover and churn tailing off as rates increase. Areas with high turnover and churn, but relatively low net migration rates include the university towns of Oxford

Figure 5

Net migration rate against churn rate for all districts in Great Britain, 2000–01

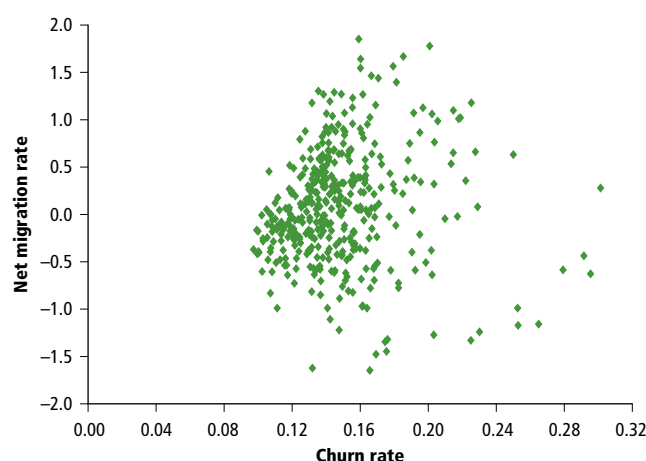
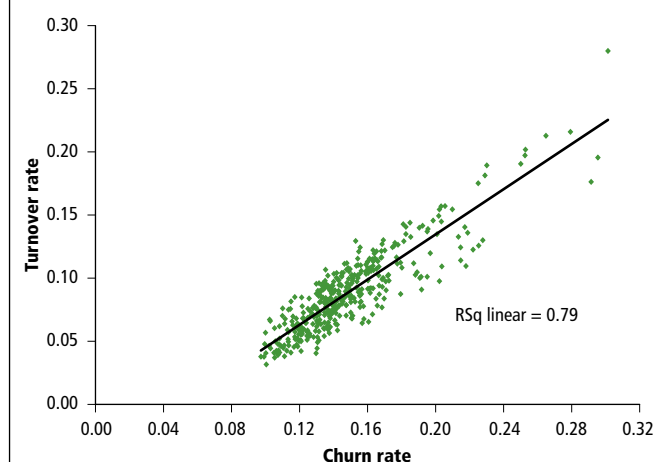
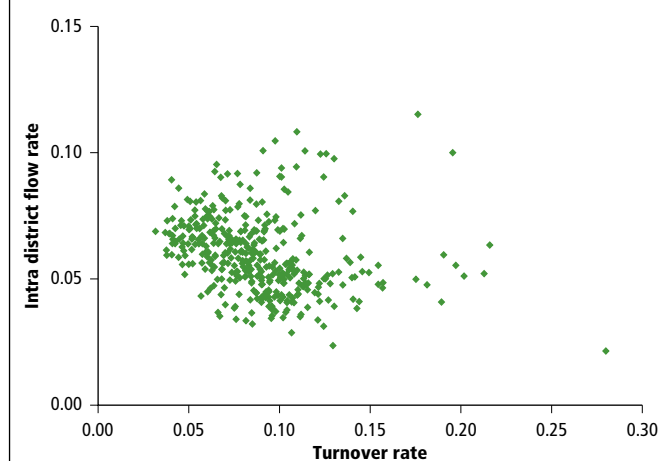


Figure 6**Turnover rate versus churn rate, districts in Great Britain, 2000–01**

and Cambridge, as well as inner London boroughs such as Islington, Wandsworth, Hammersmith and Fulham, Camden and Westminster. In these areas, while the populations may appear relatively stable when examining the low net migration figures, they actually exhibit comparatively unstable populations when the volume of moves is taken into consideration.

In order to gain an appreciation of population stability, is it important to study both turnover and churn together, or does the measurement of one reveal something about the other? **Figure 6** plots rates of turnover against churn and reveals a linear relationship between the two. The outliers represent districts with high scores for both measures of population (in) stability. All of the numbered outliers are located in central London, with the exception of 54 and 251 which are Oxford and Cambridge. Much of this linear relationship, however, is likely to be due to in-migration and out-migration being included in both turnover and churn calculations. It is well-known that area in-migration and out-migration rates are strongly correlated (Cordey-Hayes and Gleave, 1975; Rogers, 1978) although the relationship in 2001 was much stronger for whites than for non-whites (Stillwell and Hussain, 2008).

So with **Figure 6** suggesting a strong relationship between turnover and churn we may think the use of both measures is unnecessary; however, examination of **Figure 7** provides counter evidence. Here the graph plots intra-district migration rates against turnover and presents a random

Figure 7**Intra-district flow rate versus turnover rate, districts in Great Britain, 2000–01**

distribution. This suggests that there is no relationship between the rates of migration within and between districts, thus justifying the use of both turnover and churn in this analysis.

Turnover and churn rates by district type

Comparison of the aggregate population turnover and churn statistics with the net migration statistics for districts in Britain shows that these additional measures of population interaction reveal something about the stability of population that cannot be inferred from the standard net migration figures.

Table 2 gives a comparison of these statistics for total migrants as well as for males and females by the Vickers et al. family, group and class of district.

Table 1 has already shown us that the highest net migration rates for district families in Britain are Urban London and Rural UK, with considerably more people moving in than out, and out than in respectively for these two families. As was also noted, examination of the large aggregate numbers of in-migrants and out-migrants would appear to confirm the importance of Urban London and Rural UK in the internal migration story. However, as far as population stability is concerned, the size of the underlying PAR for Rural UK in comparison with the other district families means that, despite the very large volume of in-migrants and the correspondingly high in-migration rate, the populations of areas within Rural UK are relatively stable compared to other families. Looking at the aggregate population turnover and churn statistics for the four families and comparing them to both the averages for all families, groups and classes and each other, it is clear to see that Rural UK has much greater stability.

From an examination of **Table 2**, it is apparent that Urban London remains a very important location for internal migration in Great Britain, when turnover and churn are taken into consideration along with net migration. With scores of 63 and 151 persons per 1,000 for turnover and churn respectively, Urban London has the second highest turnover score for families and the highest churn score, indicating that not only does it have very significant net out-migration, but with high turnover the change in population is also significant – or at least more significant – than for all other families except Prosperous Britain. The high level of churn suggests that movement of population within districts in Urban London is also more significant than it is for other family categories. Taking all of these measures into consideration leads to the conclusion that Urban London has the most dynamic population of all families in the Vickers et al. classification.

In contrast to Urban London, Rural UK shows more stability at this aggregate level. The figures for turnover and churn for this family are 43 and 120 persons per 1,000 respectively. Not only are these significantly lower than the equivalent figures for Urban London, but they are also lower than the figures for Prosperous Britain (with scores of around 67 and 139 persons per 1,000 for turnover and churn respectively). This means that, despite far fewer people moving in and out of districts in Prosperous Britain, the movement is more perturbing than it is for Rural UK. Urban UK, on the other hand, as well as having the lowest rate of net migration, also has the lowest levels of population turnover: a rate of around 38.5 people per 1,000 of population. Levels of population churn are higher than they are for Rural UK but still lower than the mean for all families, groups and classes of district. The districts classified as Urban UK, therefore, can be seen to have far more settled or stable populations than the rest of Great Britain.

Differences in rates by gender

It can be seen in the comparison of nearly all family, group and class categories that males generally have higher rates of population turnover and churn than females, but lower rates of net migration (both in and out). This may seem counter-intuitive and needs explanation. Turnover

Table 2

Comparison of migration, turnover and churn by district classification and by sex, Great Britain, 2000–01

District Classification (Family, Group, Class)	Total net migration rate	Total turnover rate	Total churn rate	Male net migration rate	Male turnover rate	Male churn rate	Female net migration rate	Female turnover rate	Female churn rate
A: Urban UK	-0.2	39	129	-0.3	40	130	-0.1	37	127
A1: Industrial Legacy	-1.0	38	111	-1.4	39	111	-0.5	36	110
A1a: Industrial Legacy	-1.0	38	111	-1.4	39	111	-0.5	36	110
A2: Established Urban Centres	-1.7	44	126	-1.8	46	128	-1.5	42	125
A2a: Struggling Urban Manufacturing	-2.8	52	124	-2.9	55	125	-2.7	50	122
A2b: Regional Centres	2.0	95	185	2.6	100	190	1.4	91	180
A2c: Multicultural England	-3.1	48	127	-3.6	49	128	-2.7	46	127
A2d: M8 Corridor	-0.3	35	106	-0.6	38	108	0.0	33	104
A3: Young and Vibrant Cities	3.4	85	180	3.7	88	184	3.1	82	177
A3a: Redeveloping Urban Centres	3.9	81	174	4.0	84	177	3.8	78	170
A3b: Young Multicultural	1.6	109	209	2.6	112	213	0.7	106	206
B: Rural UK	2.7	43	121	2.6	45	123	2.8	42	119
B1: Rural Britain	3.6	65	133	3.6	68	135	3.5	63	130
B1a: Rural Extremes	0.7	65	135	1.0	67	137	0.4	63	133
B1b: Agricultural Fringe	4.3	76	138	4.3	79	141	4.3	73	135
B1c: Rural Fringe	4.1	83	141	4.1	87	144	4.1	79	137
B2: Coastal Britain	6.6	62	138	6.9	67	142	7.0	62	137
B2a: Coastal Resorts	7.3	84	160	7.8	87	164	6.8	81	157
B2b: Aged Coastal Extremities	5.2	60	134	5.3	66	138	6.3	62	134
B2c: Aged Coastal Resorts	10.6	87	144	11.5	91	148	9.8	83	141
B3: Averageville	-0.2	60	124	-0.8	64	127	-0.2	59	124
B3a: Mixed Urban	-0.4	62	121	-1.2	66	124	-0.4	62	121
B3b: Typical Towns	0.1	68	135	-0.1	70	137	0.2	65	133
B4: Isles of Scilly	17.8	135	201	19.7	138	202	15.9	132	200
B4a: Isles of Scilly	17.8	135	201	19.7	138	202	15.9	132	200
C: Prosperous Britain	-0.5	67	139	-0.2	69	142	-0.8	64	136
C1: Prosperous Urbanities	0.6	102	170	1.1	106	174	0.1	98	166
C1a: Historic Cities	3.5	106	175	3.9	110	179	3.1	102	171
C1b: Thriving Outer London	-2.4	107	169	-1.8	111	174	-3.0	102	164
C2: Commuter Belt	-1.1	76	137	-0.9	79	140	-1.3	73	134
C2a: The Commuter Belt	-1.1	76	137	-0.9	79	140	-1.3	73	134
D: Urban London	-8.5	63	151	-8.2	65	154	-8.9	61	149
D1: Multicultural Outer London	-8.0	92	150	-7.5	95	152	-8.5	90	148
D1a: Multicultural Outer London	-8.0	92	150	-7.5	95	152	-8.5	90	148
D2: Mercantile Inner London	-10.2	147	225	-11.1	151	229	-9.4	144	221
D2a: Central London	-10.3	147	225	-11.3	151	229	-9.3	144	221
D2b: City of London	2.8	280	301	16.9	276	295	-13.4	284	309
D3: Cosmopolitan Inner London	-8.2	111	178	-7.3	115	182	-9.1	108	174
D3a: Afro-Caribbean Ethnic Borough	-6.1	127	190	-4.7	131	195	-7.3	124	186
D3b: Multicultural Inner London	-11.8	118	174	-11.2	119	174	-12.0	114	170

and churn are measures that take into account **total** population movements in relation to the underlying population at risk in a way that net migration does not. Net migration will only indicate the balance of movement (either in or out) in relation to the population; this shows if an area is gaining or losing population, and the relative level of this gain or loss. Population turnover and churn will not give an indication of the balance of movement, but will give a standardised measure of the **amount** of movement in relation to the population at risk. Higher levels of turnover and churn mean that there will be greater numbers of people moving **in total** in relation to the underlying population, whereas higher levels of net migration just show that there are more people moving in a particular direction. The evidence here suggests that when females move in or out of family, group or class categories, the balance of movement leans more heavily to either in or out. The direction of flow is more asymmetric, but total turnover and churn rates are comparatively low. The net rate of male movement is higher (in the context of nearly every district type), but these movements are more balanced in either direction than those of females. In other words, the female population is being redistributed around the country through net migration at a rate that is

faster than for males. One explanation for this may well be associated with the greater number of females than males in the student population.

Focusing on specific groups and classes, it becomes clear that (ignoring the Isles of Scilly and the City of London) classes within Urban London, as well as those urban areas with large student populations and dynamic economies (Regional Centres, Young Multicultural, Redeveloping Urban Centres, Historic Cities) have the highest levels of population turnover and churn and correspondingly least stable populations. Groups and classes within the Rural UK family have relatively low levels of turnover and churn, with districts within the Industrial Legacy and M8 Corridor classes having the lowest levels overall, thus signifying relatively stable populations.

Standardising for age

Age has a significant influence on migration behaviour, as shown in Figure 3, with propensity to migrate reaching a peak at around the 20–24 age group. So far, however, turnover and churn statistics have not taken into consideration the effect of age. It is to be expected that

Table 3

Age standardised turnover and churn for the families in Great Britain, 2000–01

	Total Turnover rate		Total Churn rate	
	Standardised by original age groups	Standardised by 15 year age groups	Standardised by original age groups	Standardised by 15 year age groups
A: Urban UK	37.0	37.3	124.4	125.2
B: Rural UK	46.8	46.1	128.9	127.4
C: Prosperous Britain	67.6	67.4	140.2	139.9
D: Urban London	54.4	55.7	129.5	133.7

the population compositions of districts and families, groups and classes of district will differ. Urban London, for example, may have a younger age structure than Rural UK and this will necessarily affect the rates of migration associated with these areas.

To deal with these population composition effects it is possible to standardise turnover and churn rates by the age structure present in the total population. A method of 'direct standardisation' is proposed by Rowland (2006), and is used here to produce the standardised rates of turnover and churn shown in **Table 3**. Following the notation proposed by Rees et al. (2000) and taking the standardised turnover rate for area i , STO_i , as the example:

$$STO_i = \frac{\sum_a (to_i^a P_i^a)}{P_i^{as}} \quad (4)$$

where the turnover rate for age group a in zone i is:

$$to_i^{as} = \left(\frac{o_i^a + D_i^a}{P_i^{as}} \right) 1000 \quad (5)$$

where:

$$O_i^a = \sum_j M_{ij}^a$$

= the total outflows in age group a from zone i to other zones (6)

$$D_i^a = \sum_j M_{ji}^a$$

= the total inflows in age group a to zone i from other zones (7)

and

$$P_i^a = \sum_j P_{ij}^a$$

= the population in age group a in all zones in the system (8)

Calculating the age-standardised rate of churn follows the same form but with the addition of the term W_i^a to represent within-zone moves so that the churn rate for age group a in zone i is:

$$STCH_i = \frac{\sum_a (ch_i^a P_i^a)}{\sum_a P_i^{as}} \quad (9)$$

where the churn rate for age group a in zone i is:

$$ch_i^{as} = \left(\frac{o_i^a + D_i^a + W_i^a}{P_i^{as}} \right) 1000 \quad (10)$$

This method of direct standardisation produces the set of age standardised rates shown in Table 3. For both turnover and churn, the rates have undergone two methods of standardisation. In the first method, the rates were standardised by the age groups contained in the original data. In the second, the rates were standardised by the 15 year age groups used in the majority of this analysis. These new rates allow for the effect of differing age structures between the families. A comparison with rates in Table 2 indicates some age effects, although these are more pronounced with churn than turnover. Taking turnover first, the relationships between the families do not change. Highest rates of turnover are found in Prosperous Britain and Urban London, with the lowest rates found in Urban UK and Rural UK. The importance of Prosperous Britain and Rural UK increases slightly, whilst for Urban UK and Urban London, the importance decreases – in the case the latter

by a relatively large margin. There is little difference between the two methods used to standardise turnover.

In contrast, age standardisation affects the churn rates more noticeably, especially for Urban London. Whereas the non-standardised rates for Urban London show the highest rates of churn, the standardised rate is less than that computed for Prosperous Britain. This suggests that much of the churning of the population in London could be attributed to a younger age structure. Similarly, whereas Rural UK experiences the lowest rates of churn with the non-standardised calculations, standardised rates are higher than for Urban UK, suggesting conversely that the lower degree of churning could partially be attributed to an older age structure.

Although standardising turnover and churn rates for age produces a small caveat in relation to the interpretation of the rates in general, the standardisation does not dramatically alter turnover and churn statistics for district families in most cases. Where differences are more pronounced, as in the case of Urban London, it suggests that much of the population instability demonstrated here is because of the movement of a young, labour-force-age population. While significant, this does not necessarily mitigate the importance of the characteristics of the Urban London family in explaining migration patterns, but rather helps stress the interactions between age/life-course and the environmental, social and economic characteristics of places. The maps shown in **Figures 8.1–8.6** reveal exactly how these general trends of turnover and churn present themselves spatially for the five main age groups used in this study.

Patterns of turnover and churn by district

In terms of the aggregate patterns of turnover and churn, studying the figures in relation to the district classification has already shown that highest levels of turnover and churn are found in London and some of the more dynamic urban areas in Britain. It is also noticeable that other areas of relatively high turnover and churn tend to be more concentrated in the South East, around London, and towards the South West. This is evident on both the 'all ages' maps shown in Figure 8. Areas defined as Industrial Legacy and M8 Corridor have particularly low levels of turnover and churn, and these maps show areas close to South Wales, Birmingham, Manchester, Liverpool, West and South Yorkshire, the North East and the M8 corridor between Glasgow and Edinburgh as fitting into the patterns shown in the classification analysis.

It is evident that the gap between areas of lowest turnover and other higher turnover areas (in terms of numbers of people moving) is proportionally greater than the same gap in churn statistics for the same areas. Put another way, areas of low turnover also have low churn, but the lower turnover is more noticeable than the lower churn when compared with other higher scoring areas. The range between lowest and highest is greater for turnover than churn. As churn takes into account intra-district migrations and turnover does not, it can be concluded that, compared with districts in the rest of the country, when migrations occur in these low turnover/churn areas, they are more likely to be shorter-distance, intra-district migrations than longer-distance, inter-district migrations. This is confirmed by looking at the average relationship between turnover and churn for all districts (**Table 4**). On average, there are around 60 more people per 1,000 moving per district when intra-district migrations are taken into account. Districts where the figure is greater than average experience relatively more intra-district migration. The gap between turnover and churn for the majority of the low turnover districts (those in South Wales, the North East, the Industrial North) is greater than 60. In these areas of low turnover and churn, if migrations take place they are more likely to be local, within-district movements – movements less likely to have a perturbing effect on population stability.

The remaining maps (Figures 8.1 to 8.6) show the relative levels of turnover and churn for other key age groups, with (as expected) the

Figure 8.1

Turnover rates by district, all ages, Great Britain, 2000–01

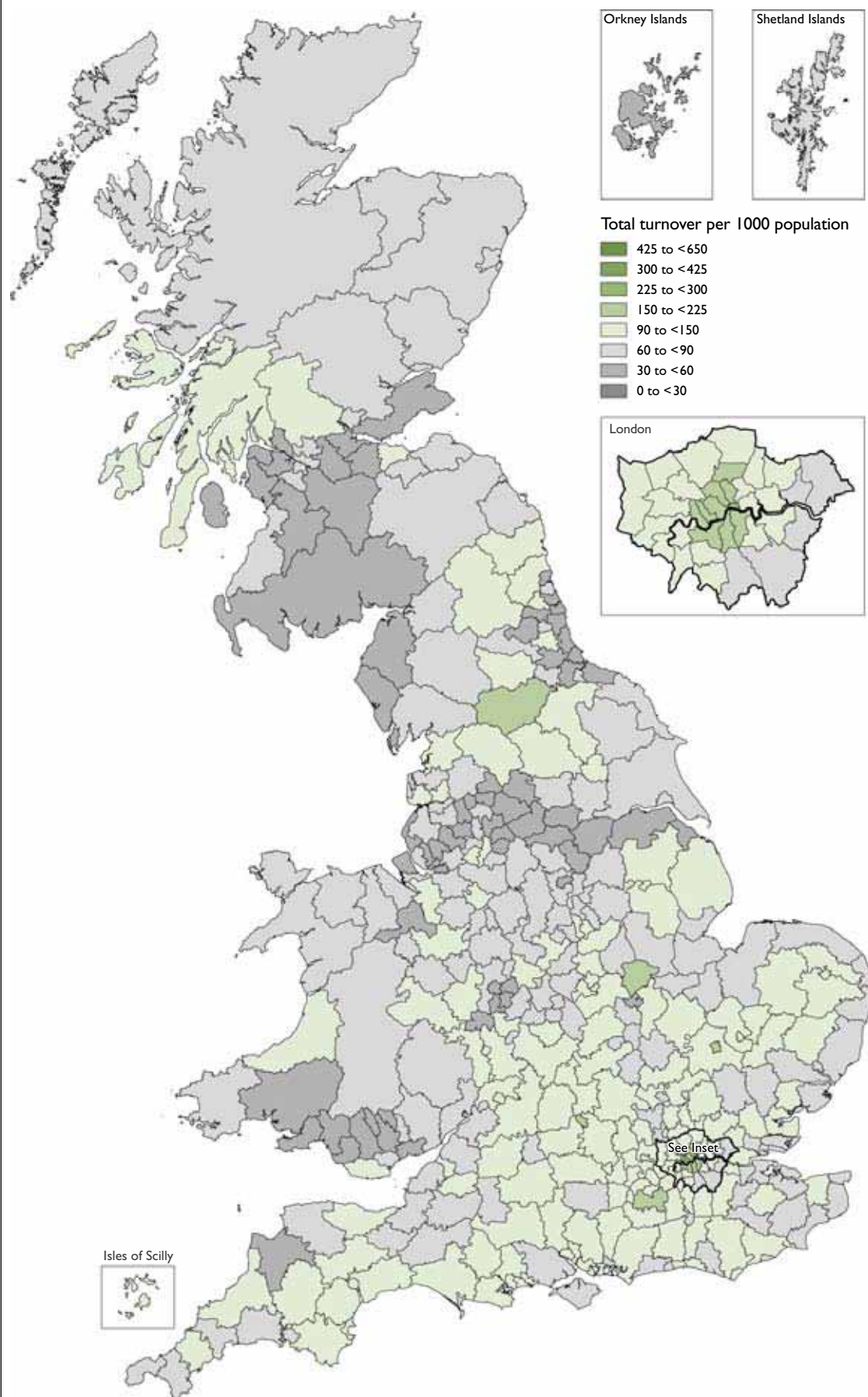


Figure 8.2

Churn rates by district, all ages, Great Britain, 2000–01

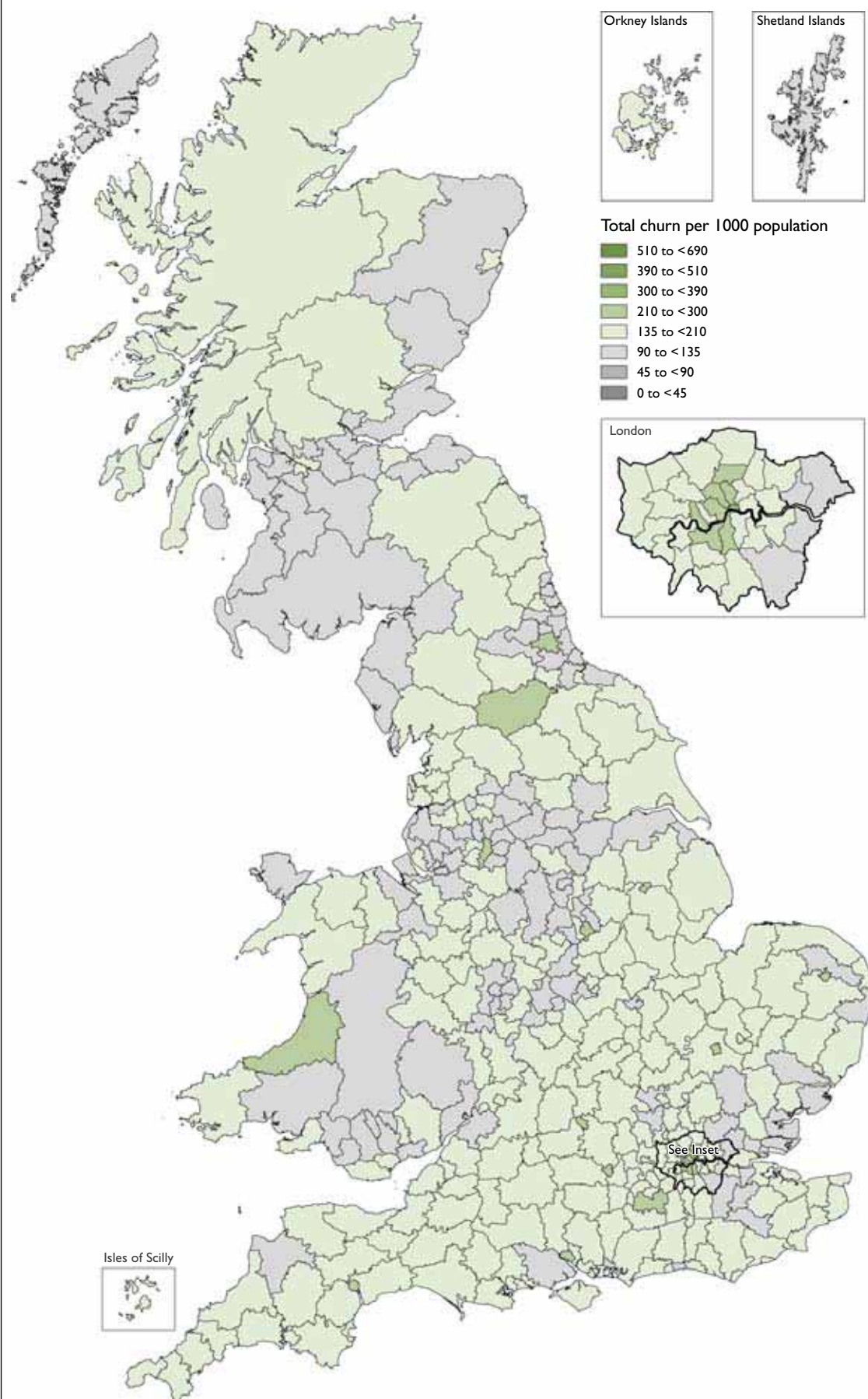


Figure 8.3

Turnover rates by district, 16–29 years, Great Britain, 2000–01

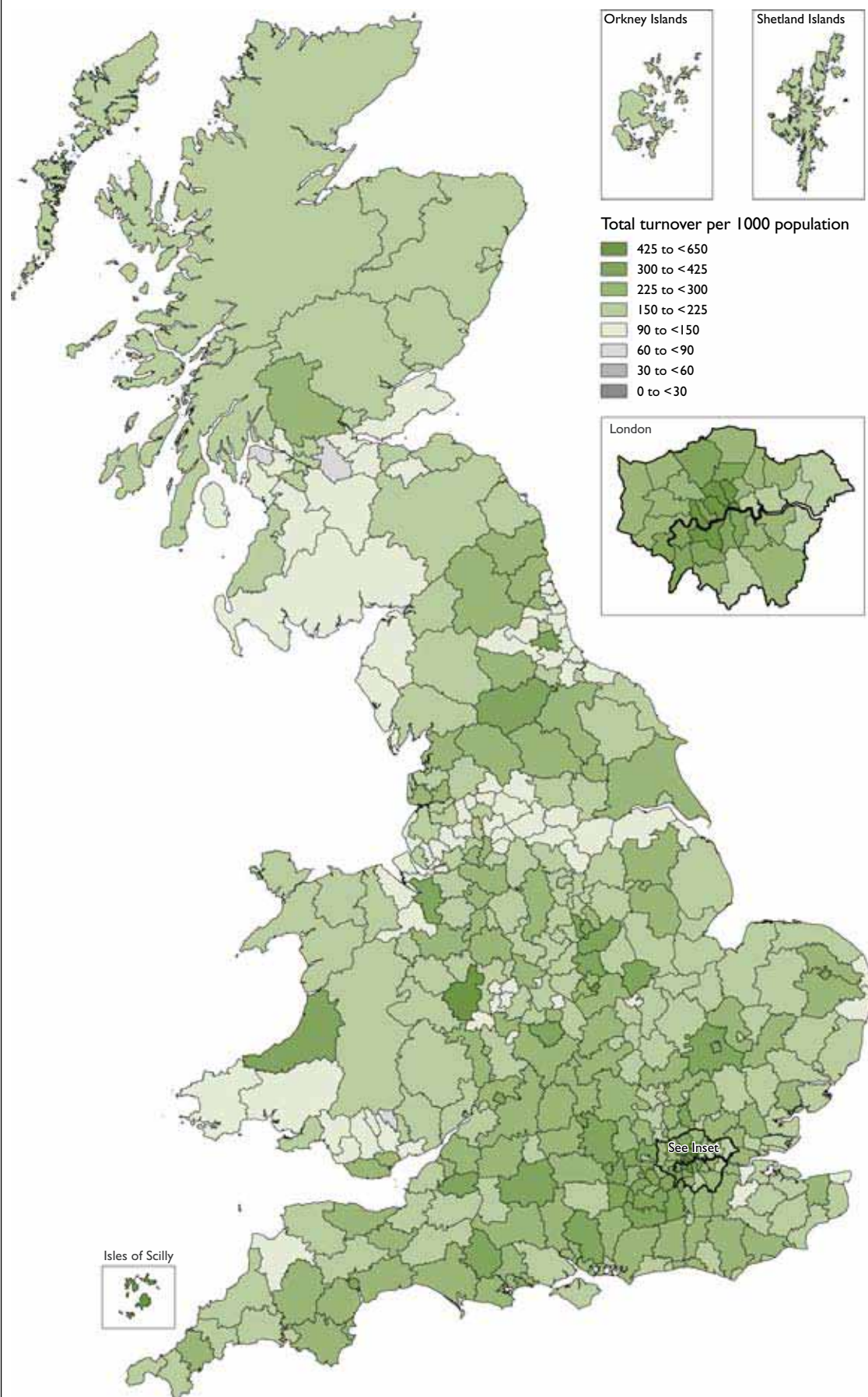


Figure 8.4

Churn rates by district, 16–29 years, Great Britain, 2000–01

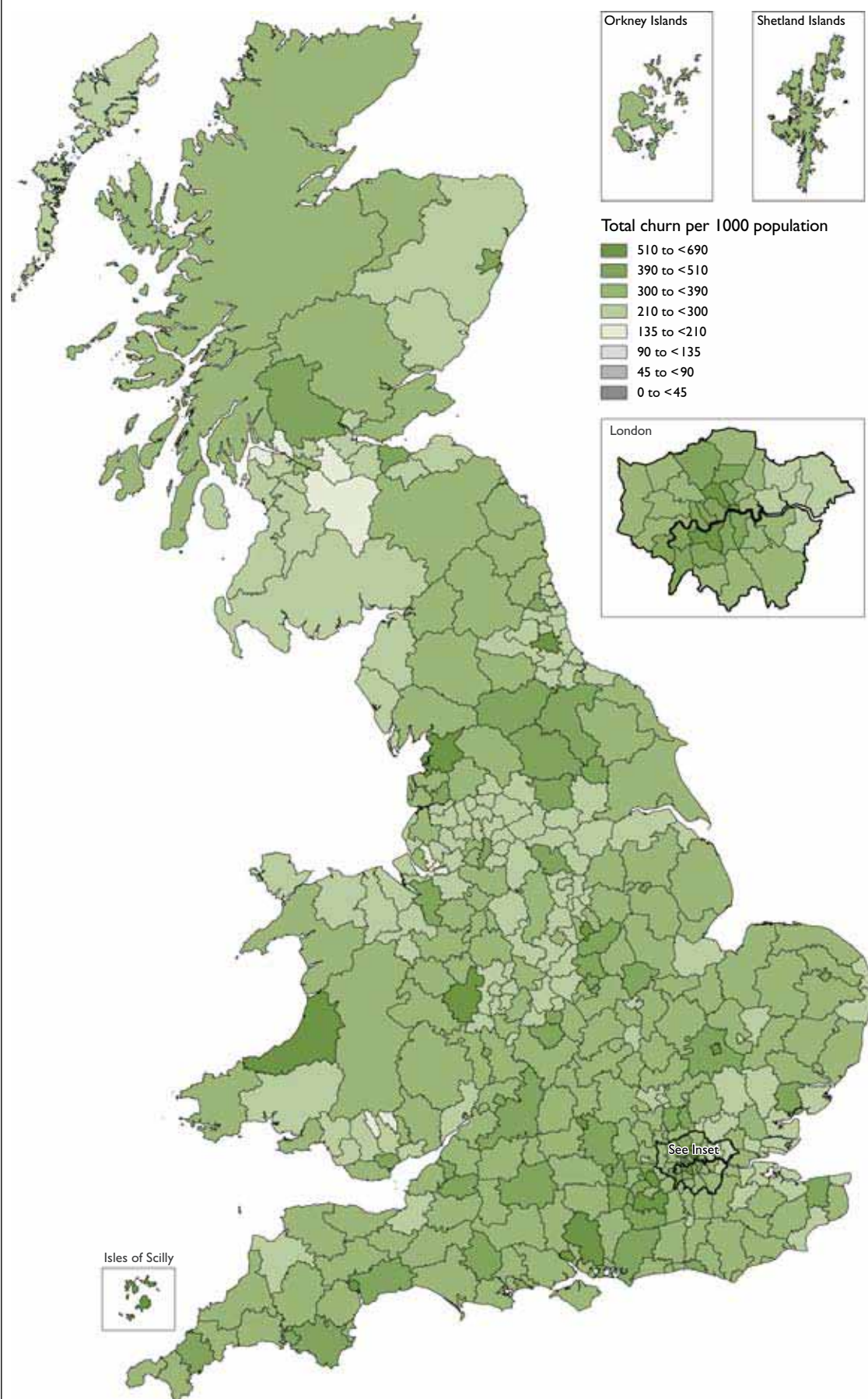


Figure 8.5

Turnover rates by district, 45 years – pensionable age, Great Britain, 2000–01

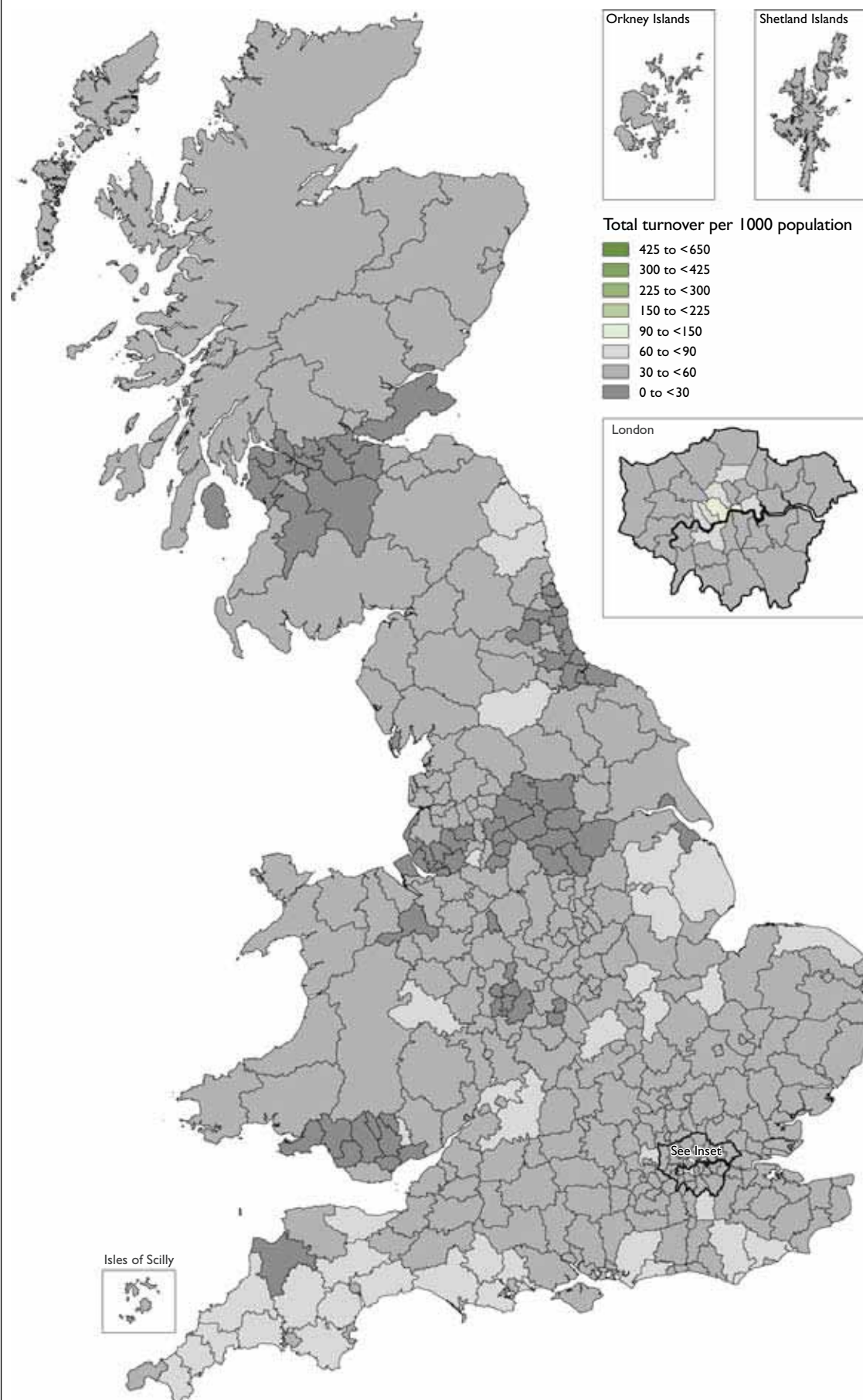


Figure 8.6

Churn rates by district, 45 years – pensionable age, Great Britain, 2000–01

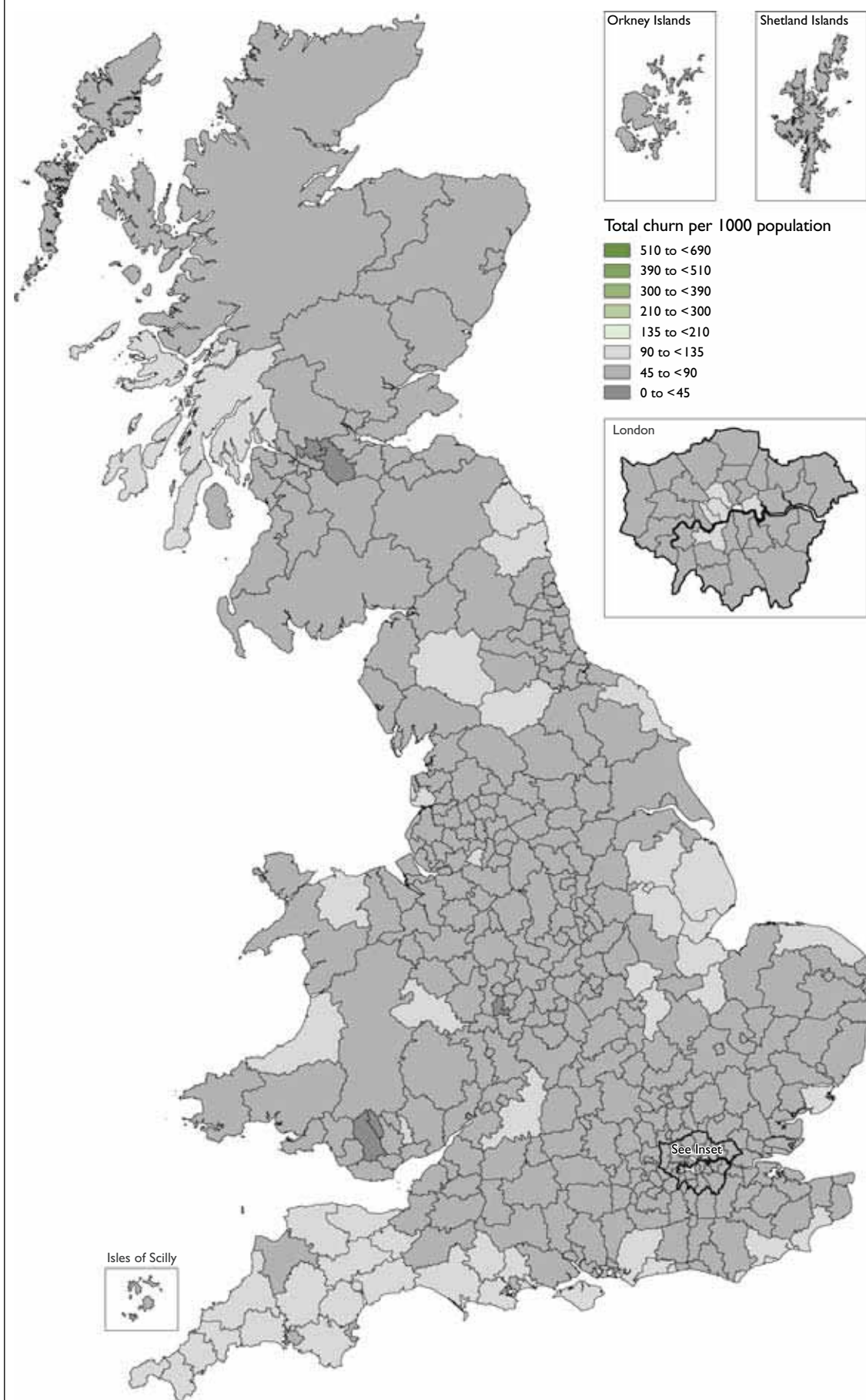


Table 4

Summary statistics for population turnover and churn for districts in Britain: by broad age group and by breakdown of the 16–29 age group, 2000–01

Summary turnover statistics for individual GB districts

	Total	Male	Female	0–15	16–29	30–44	45–PA	Pensionable age	16–17	M16–17	F16–17	18–19	M18–19	F18–19	20–24	M20–24	F20–24	25–29	M25–29	F25–29
Max	280	276	284	221	614	300	120	67	615	667	545	818	750	1000	743	749	847	572	549	598
Min	32	33	31	29	76	35	14	11	23	22	19	56	37	77	88	79	90	77	77	76
Mean	88	92	85	70	217	97	44	31	64	64	63	254	230	278	275	262	287	211	218	203
Median	85	88	82	70	206	95	44	32	57	54	58	237	211	267	264	251	275	197	204	191
Range	248	243	254	192	538	265	106	57	592	644	527	762	713	923	655	670	757	496	472	522

Without Isles of Scilly/City of London

Max	216	225	213	221	460	211	97	57	292	421	211	641	617	667	638	749	689	433	462	437
Min	32	33	31	29	76	35	14	11	23	22	19	56	37	77	88	79	90	77	77	76
Mean	88	91	84	70	215	96	44	31	62	62	61	251	228	274	273	261	285	210	217	202
Median	85	88	82	70	205	95	44	32	57	54	58	236	209	266	263	251	275	196	204	191
Range	184	192	183	192	384	177	83	46	269	398	192	585	579	589	550	670	599	356	385	360

Summary churn statistics for individual GB districts

	Total	Male	Female	0–15	16–29	30–44	45–PA	Pensionable age	16–17	M16–17	F16–17	18–19	M18–19	F18–19	20–24	M20–24	F20–24	25–29	M25–29	F25–29
Max	301	302	309	299	639	326	141	88	615	667	545	909	750	1333	819	796	966	597	563	633
Min	97	97	97	101	182	104	42	32	62	57	65	137	97	180	207	173	240	193	184	202
Mean	148	151	145	141	334	164	72	55	116	109	123	349	300	400	424	392	457	333	342	325
Median	142	145	139	139	320	163	70	54	110	102	118	329	279	379	414	379	448	326	336	317
Range	204	205	212	198	457	222	99	56	553	610	480	772	653	1153	612	623	726	404	379	432

Without Isles of Scilly/City of London

Max	295	302	289	299	617	279	128	88	347	449	267	703	668	857	819	796	851	553	548	559
Min	97	97	97	101	182	104	42	32	62	57	65	137	97	180	207	173	240	193	184	202
Mean	148	150	145	141	333	164	72	55	114	108	121	347	299	396	422	390	454	333	341	325
Median	142	144	139	139	320	162	70	54	110	101	117	329	278	379	413	378	447	326	335	317
Range	198	205	192	198	436	175	86	56	284	392	202	567	571	677	612	623	611	360	364	357

highest levels in the 16–29 age group and the lowest levels in the ‘pensionable age and above’ age group (not shown). Two points can be made in summary. Firstly, while there are no immediately apparent spatial patterns in the 16–29 age group, closer inspection of the maps reveals the highest levels of turnover and churn are in the spatially diffuse but characteristically homogenous (as confirmed by the Vickers et al. classification) urban areas previously described – the dynamic, growing university towns and urban areas fringing London.

Secondly, there is more clarity in the spatial patterning related to higher levels of mobility in the groups with lowest overall levels of turnover and churn (the two oldest age groups – only ‘45 to pensionable age’ is shown), with coastal areas – the South West in particular, showing higher levels of these measures of stability for these older groups. Low levels of turnover and churn are apparent for the ex-industrial areas identified in the aggregate analysis, for all age groups. These low levels are, however, particularly pronounced at the family-centred age ranges (0–15 and 30–44 – not shown, although the patterns for these age groups are almost identical to the ‘all ages’ patterns), less so with the other age ranges.

Table 4 provides summary statistics for the main age groups, as well as a more detailed breakdown for the 16–29 age group. Of note here (as with net migration) are the continually higher mean rates of population turnover and churn for females when compared to males for the post–16 age groups. This difference is especially pronounced when looking at the late-teen and early-twenties age ranges. Aside from this difference, the highest mean rates of turnover and churn for both males and females are found in the 20–24 year old age group, and are, respectively, around 70 and 121 migrants per 1,000 people higher than the averages for the 16–29 age group.

Conclusions

The understanding of internal migration in Great Britain has been advanced further by the introduction of turnover and churn analysis. These two measures help to quantify the stability of a population in an area; the latter developing the concept of stability slightly further than the former by taking account of intra-zonal as well as inter-zonal flows. The definition of stability we use here is based on the proportion of residents living at an address in an area who remain there from one year to the next. Generally speaking, the areas with the least stable or most transient populations are urban areas. The most stable or least changing populations are found in rural and previously industrial areas. Levels of stability vary greatly between different age groups with the 16–29 age group (and more specifically within that group the 18–19 and 20–24 age ranges) being the least stable, and the older age groups being inherently more stable. Within each age group though there are specific areas within Great Britain that have more or less stable sections of these populations.

Analysis of turnover and churn statistics for Great Britain at the Vickers et al. classification family level has shown that while London retains the importance it has when examining net migration balances and rates, the role of rural Britain becomes less important, even when the distorting effect of differential age profiles across the district classifications is taken into consideration. The vast size of the underlying rural population in Britain means that despite the apparently large net in-migration to rural areas, the disturbing effect that this has on the resident population is relatively low. Rural areas tend to have, on the whole, much more stable populations than areas in Prosperous Britain, though the naming of Prosperous Britain may in this case be misleading.

Key findings

- Turnover and churn statistics enhance our understanding of the migration processes that affect population stability though providing information on the volume of flows experienced by different areas. There is no clear relationship between net-migration rates and turnover and churn statistics
- Population stability varies considerably with age with the population of people in their late teens and early twenties being least stable and older age groups being more stable
- Rural Britain, whilst important in the net-migration story (exhibiting considerable in-migration), becomes less important in terms of population stability and is relatively stable due to a large underlying population
- Ex-industrial urban Britain also features a relatively stable population, while other urban areas including London exhibit more transient, unstable populations, although in London this is partially attributable to a younger migrant age structure
- Age standardisation reveals that the relative instability of the population in London is partially attributable to the young age structure of migrants. Areas surrounding London in Prosperous Britain exhibit even less stability when age is taken into account.

Commuter Belt makes up much of Prosperous Britain, and can in many ways be regarded as rural. There appears to be continuation of the trend recognised by Rees et al. (1996) of Urban London populations with the relatively footloose Urban London migratory characteristics, occupying space that would otherwise be recognised as rural. In effect, it could be argued that there is an identifiable two-tier 'rural' in Britain. The 'traditional rural' Britain, with a generally stable population, perhaps experiencing some in-migration from more classically urban areas, and the 'new rural' Britain, which may outwardly exhibit many of the same environmentally rural characteristics as the traditional rural Britain, but that features this 'prospering' population with some of the migratory characteristics associated with the population of Urban London. To complete the narrative at this family level, turnover and churn statistics have helped reinforce the idea of Urban UK (which should perhaps more accurately be described as 'ex-industrial urban Britain') as being an area where populations are more stable; migrations, if they do happen, tend to be short distance; longer-distance migrations are more rare.

As numerical expressions of population stability, turnover and churn variables are valuable and could be put to further use in subsequent analyses of internal migration. A comparison of turnover rates between 2001 Census and patient register data at the district scale in England and Wales shows a reasonable level of correlation (correlation coefficient of 0.73). Preliminary analysis of turnover rates based on patient registration over time between 2001 and 2006 suggests that it is rural areas such as Conway, Anglesey and Gwynedd where turnover has declined most significantly (by over 20 per cent) whereas turnover has increased most appreciably in urban areas such as Barking and Dagenham, Newham and Corby. Furthermore, turnover and churn variables could be incorporated into mathematical or statistical explanatory models, or included in the development of an alternative area-classification framework, based entirely around migration variables. Such a framework would be very useful in monitoring change in migration at the district level beyond 2001 and is part of our ongoing research.

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